2. The radius of the circle is 3.12 m. Calculate the area of the circle with regard to significant figures. Ans: (30.6 m^2)

3. Assuming that the frequency γ of a vibrating string may depend upon i) applied force (F) ii) length (ℓ) iii) mass per unit length (m), prove that $\gamma \alpha \frac{1}{l}\sqrt{\frac{F}{m}}$ using dimensional analysis. (related to JIPMER 2001)

4. Jupiter is at a distance of 824.7 million km from the Earth. Its angular diameter is measured to be 35.72″. Calculate the diameter of Jupiter. Ans: (1.428×10^5 km)

5. The measurement value of length of a simple pendulum is 20 cm known with 2 mm accuracy. The time for 50 oscillations was measured to be 40 s within 1 s resolution. Calculate the percentage accuracy in the determination of acceleration due to gravity 'g' from the above measurement. Ans: (6%)

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UNIT-1(XI-Physics_Vol-1).indd 39 05-01-2022 22:12:08